FINS5545 Project Part A: Data Foundation

This Part A draft builds the data foundation for a systematic multi-asset fund app. The target user is an investor or analyst who wants transparent equity, crypto and news inputs before moving to portfolio construction and sentiment analytics in Part B.

Data sources and cleaning

The project loads all raw data through the provided data_access helper. Equity and crypto prices are sorted by ticker and date, duplicate ticker-date rows are removed, and only positive adjusted close observations through 2023-12-31 are retained. News is de-duplicated by ticker, date and title.

Dataset	Rows	Tickers	Dates	Duplicate keys
equity_prices	50,300	50	2020-01-02 to 2023-12-29	0
crypto_prices	14,610	10	2020-01-01 to 2023-12-31	0
news_headlines	146,836	50	2020-01-01 to 2023-12-31	0

Return features

Returns are simple daily percentage returns computed within each ticker before cross-asset alignment. This prevents crypto weekend observations from contaminating equity trading-day calculations. Outliers are retained because they are real market moves and are important for risk measurement.

Asset	Obs.	Mean	Std	Min	P99	Max
Crypto	14,600	0.223%	5.199%	-42.96%	16.12%	74.92%
Equity	50,250	0.059%	2.304%	-52.01%	6.58%	33.70%

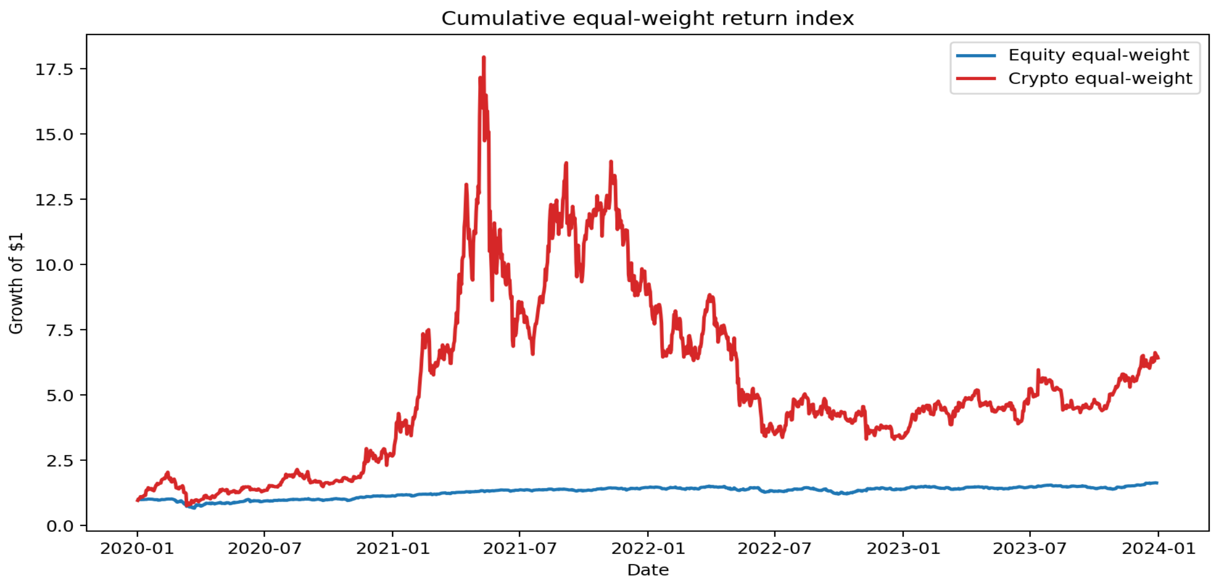

Figure 1. Cumulative equal-weight return indices for equities and crypto.

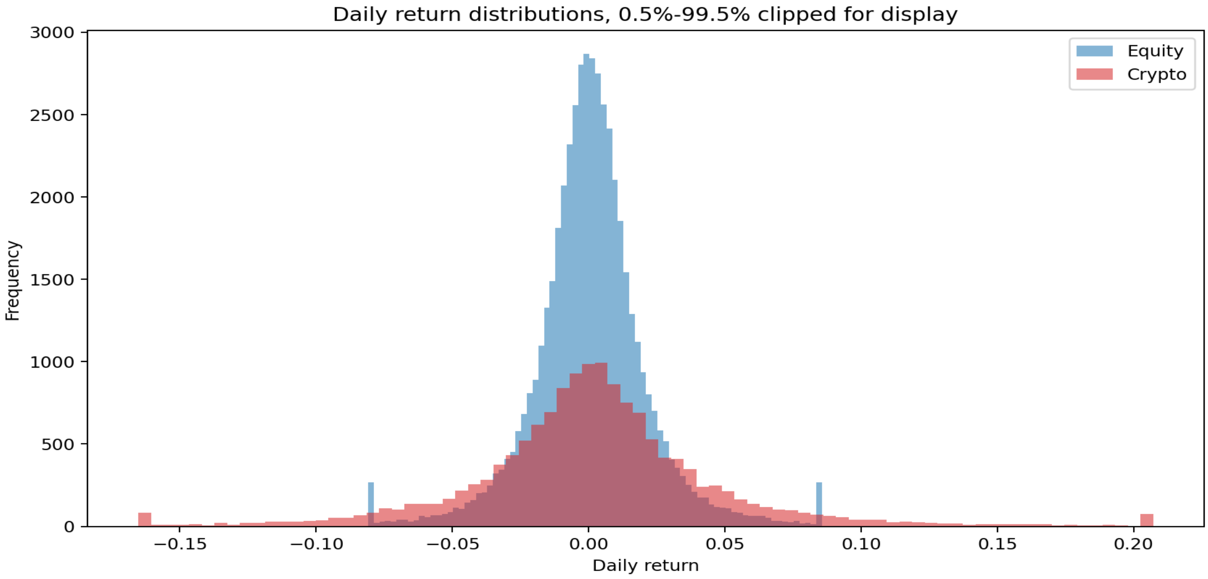

Figure 2. Daily return distribution with display clipping; original outliers are retained in the data.

Text panel

Headlines are assembled into a ticker-sector daily text panel. Weekend and non-trading-day headlines are mapped to the next equity trading day. Part A deliberately does not score sentiment; it only reports article counts, word counts and signal-word counts to describe coverage.

The largest news sector by article count is Tech with 26,632 articles. These coverage differences matter because Part B sentiment signals must account for sparse or uneven news flow.

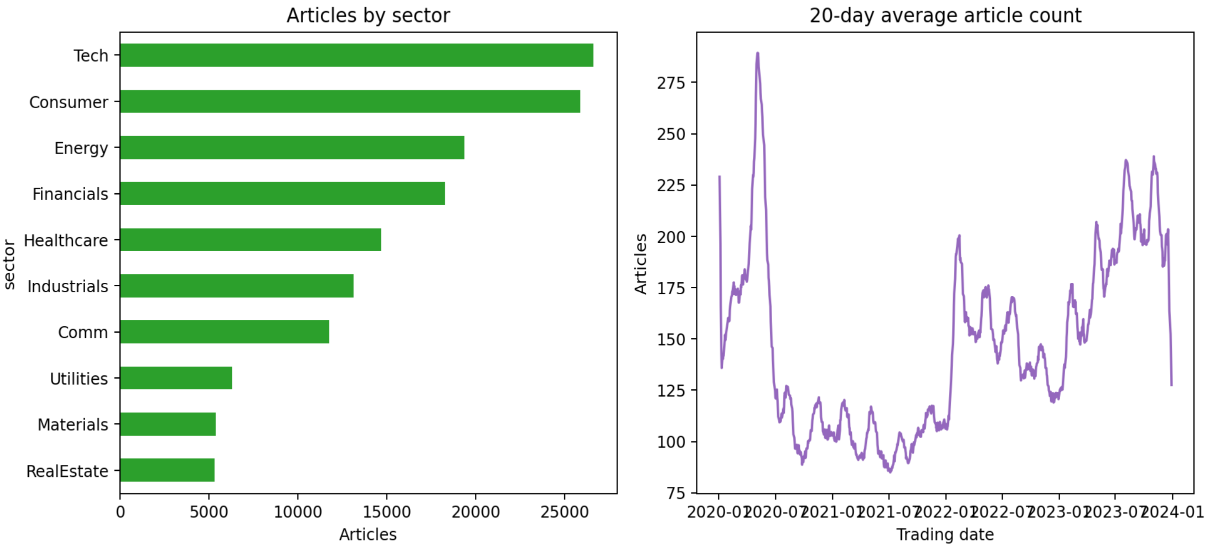

Figure 3. Sector article coverage and time-series news volume.

Next step

This foundation enables Part B to estimate walk-forward funds, compute lagged sentiment indices, and serve the precomputed outputs in a Streamlit dashboard without committing raw data.